Notes: Colacito and Croce (JPE, 2011)

Risks for the Long Run and the Real Exchange Rate

Contents

1 Big picture

- **Question.** Can a frictionless international asset-pricing model explain why exchange rates are not wildly volatile even though cross-country consumption growth is weakly correlated, and can the same model also match international stock-market comovement and equity premia?
- **Setting.** The paper builds a two-country equilibrium model with complete markets, Epstein-Zin recursive preferences, and long-run risk in consumption and dividends. It then confronts the model with U.S. and U.K. data.
- **Core puzzle.** In standard power-utility models, exchange-rate growth should equal differences in countries' stochastic discount factors. Because consumption growth is only weakly correlated across countries, these models need implausibly large risk aversion and then predict exchange-rate volatility far above what is observed.
- **Main theoretical contribution.** The paper shows that if agents care about the timing of resolution of uncertainty and if countries share highly correlated long-run growth prospects, then stochastic discount factors can be much more correlated than current consumption growth. Exchange rates then need to move much less.
- **Main mechanism.** Exchange rates respond not only to short-run consumption shocks but also to news about persistent future growth. When long-run risks are highly correlated across countries, a lot of future consumption news cancels out in relative prices.
- **Main empirical strategy.** The paper first shows that consumption data alone have very weak power to identify persistent long-run components. It then uses predictive variables from financial markets — especially price-dividend ratios, risk-free rates, consumption-output ratios, lagged consumption growth, and default premia — to construct observable measures of long-run risk in the United States and the United Kingdom.
- **Main calibration result.** A symmetric benchmark calibration with highly correlated long-run shocks can simultaneously generate realistic exchange-rate volatility, highly correlated international stock returns, persistent price-dividend ratios, and sizable equity premia.
- **Main empirical result.** The predictable components of U.S. and U.K. consumption growth are small but highly persistent and strongly correlated. Their correlation rises over time while exchange-rate volatility falls.
- **Main parameter result.** In the post-Bretton Woods sample, the estimates support an intertemporal elasticity of substitution above one and moderate risk aversion. In the richer joint estimations, the intertemporal elasticity is around two and risk aversion is well below the levels required in standard international equity-premium arguments.
- **Main limitation.** The model abstracts from stochastic volatility. As a result, it cannot address uncovered-interest-parity failures or time-varying currency premia.
- **Economic takeaway.** Common long-run growth news is a powerful way to reconcile international prices and quantities. The paper offers a clean mechanism for why exchange rates can be much smoother than short-run consumption data alone would suggest.

2 Model setup, calibration, and empirical measurement (key objects)

2.1 Two-country environment

- There are two countries, home and foreign.
- Markets are complete.
- Each country consumes only its own endowment good because the model imposes complete home bias in preferences.
- The paper is careful to note that a high cross-country correlation of pricing kernels should not automatically be read as "perfect risk sharing," because home bias breaks the direct mapping between the two ideas.

Interpretation. The model is designed to isolate the asset-pricing implications of long-run consumption risk, not to build a full production economy with trade in many goods.

2.2 Preferences and the key state variable

- Preferences are Epstein-Zin recursive preferences.
- The crucial state variable is a persistent expected-growth component x_t in consumption growth.
- Short-run shocks are $\epsilon_{c,t}$ for consumption and $\epsilon_{d,t}$ for dividends.
- Long-run news is $\epsilon_{x,t}$, the shock to the predictable component x_t.

Interpretation. In a standard CRRA setting, only current consumption growth differences matter for exchange rates. Here, future growth news matters today because recursive preferences separate risk aversion from the intertemporal elasticity of substitution.

2.3 Cross-country correlation structure

The paper distinguishes three cross-country correlations:

- correlation of short-run consumption shocks,
- correlation of short-run dividend shocks,
- correlation of long-run shocks.

The key modeling choice is that the long-run shocks are much more strongly correlated across countries than short-run consumption growth.

Interpretation. This is the heart of the mechanism. Countries may look weakly correlated at high frequencies but still share very similar long-horizon growth prospects.

2.4 Benchmark calibration

The benchmark monthly calibration reported in Table 1 uses:

- intertemporal elasticity of substitution $\psi = 2$,
- risk aversion $\gamma = 4.25$,
- subjective discount factor $\delta = 0.998$,
- persistence of the long-run component $\rho = 0.987$,
- volatility ratio of long-run to short-run consumption shocks $\varphi_e = 0.048$,
- leverage parameter for dividends $\lambda = 3$,
- cross-country correlation of long-run shocks $\rho_x^{hf} = 1$,
- cross-country correlation of short-run consumption shocks $\rho_c^{hf} = 0.3$,
- cross-country correlation of short-run dividend shocks $\rho_d^{hf} = -0.1$.

Interpretation. The calibration does not need extreme risk aversion. Instead, it leans on high persistence and high cross-country correlation of the long-run component.

2.5 Why the United States and the United Kingdom

The empirical work concentrates on the United States and the United Kingdom because:

- both countries have long and relatively reliable consumption and dividend time series,
- both were financially open for a large part of the postwar period,
- and both offer enough data to estimate low-frequency components that would be hard to identify in shorter samples.

The common annual sample runs from 1929 to 2006. For model testing, the paper gives special attention to the post-Bretton Woods sample, 1971–2006.

Interpretation. The empirical problem is not just finding a cross-country panel; it is finding a cross-country panel long enough to identify a slow-moving latent component.

2.6 Data sources

- U.S. consumption, GDP, and population are from the National Income and Product Accounts.
- U.S. stock-market returns, Treasury-bill yields, dividends, and dividend yields are from CRSP.
- U.S. CPI inflation and default premia come from the Federal Reserve Bank of St. Louis.
- U.K. post-1963 consumption data come from the U.K. Office of National Accounts; earlier data come from Mitchell (1979).
- U.K. GDP, FTSE returns, short rates, dividend yields, population, and CPI inflation come from Global Financial Data.

All empirical variables are demeaned before estimation.

2.7 Predictive variables and long-run-risk measures

The paper uses several observable predictors to proxy for long-run growth prospects:

- lagged price-dividend ratio,
- lagged risk-free rate,
- lagged consumption-output ratio,
- lagged consumption growth,
- and lagged default premium.

It also considers a VAR-based measure built from consumption growth, the price-dividend ratio, and the risk-free rate.

Interpretation. The empirical move is to let financial prices summarize information about future growth that raw consumption data cannot estimate well on their own.

2.8 Target moments and outcome variables

The paper evaluates the model against a rich set of international moments, including:

- exchange-rate volatility and autocorrelation,
- the Backus-Smith correlation between exchange-rate growth and cross-country consumption-growth differentials,
- average equity premia and risk-free rates,
- volatility and persistence of price-dividend ratios,
- cross-country correlations of returns, dividends, and price-dividend ratios,
- and foreign-currency Sharpe ratios.

Interpretation. The paper is not aiming to fit just one puzzle. It wants one mechanism to match several international prices and quantities simultaneously.

2.9 Preview facts from the empirical section

The main descriptive facts that come out of the predictive-regression exercises are:

- the long-run component explains only a modest share of annual consumption-growth variance,
- but it is very persistent,
- and its correlation across the United States and the United Kingdom can be very high,
- while the correlation appears to rise in periods when exchange-rate volatility falls.

Interpretation. The component is small in variance terms but large in asset-pricing terms because it is highly persistent and therefore matters a lot for marginal utility under recursive preferences.

3 Models and empirical specifications (all equations + interpretation)

3.1 Complete-markets benchmark and the international puzzle

With complete markets, the real exchange rate must satisfy

$$\Delta e_{t+1} = m^*_{t+1} - m_{t+1}. \tag{1}$$

Interpretation. The exchange rate is pinned down by the difference in the two countries' log stochastic discount factors. If those discount factors are not highly correlated, the exchange rate has to move a lot.

Under standard power utility, the stochastic discount factor is closely tied to current consumption growth. Because cross-country consumption growth is only weakly correlated, this benchmark implies excessive exchange-rate volatility unless one uses implausibly large risk aversion.

3.2 Epstein-Zin pricing kernel

For the home country, the pricing kernel is

$$m_{t+1} = \frac{1-\gamma}{1-1/\psi}\log\delta - \frac{1-\gamma}{\psi-1}\Delta c_{t+1} + \frac{1/\psi-\gamma}{1-1/\psi}r_{c,t+1}. \tag{2}$$

Interpretation. The stochastic discount factor depends not only on current consumption growth but also on the return to aggregate wealth. This is exactly what lets future growth news affect current valuations.

3.3 Laws of motion for consumption, dividends, and long-run risk

The model specifies

$$\Delta c_t = \mu_c + x_{t-1} + \epsilon_{c,t}, \tag{1}$$
$$\Delta d_t = \mu_d + \lambda x_{t-1} + \epsilon_{d,t}, \tag{2}$$
$$x_t = \rho x_{t-1} + \epsilon_{x,t}. \tag{3}$$

The foreign country has identical equations with starred variables, and the shocks can be cross-country correlated.

Interpretation.

- x_t is the long-run growth component.
- Consumption and dividends both load on x_t, but dividends are more levered through λ.
- Because x_t is persistent, a shock to it changes a long stream of future expected cash flows and future consumption.

3.4 Linearized pricing system

A first-order approximation delivers

$$m_{t+1} = \log \delta - \frac{1}{\psi} x_t + k_c \frac{1-\gamma\psi}{\psi(1-\rho k_c)} \epsilon_{x,t+1} - \gamma \epsilon_{c,t+1}, \tag{3}$$

$$\Delta e_{t+1} = m^*_{t+1} - m_{t+1}, \tag{4}$$

$$v_{d,t} = \bar{v}_d + \frac{\lambda - 1/\psi}{1-\rho k_d} x_t, \tag{5}$$

$$v_{c,t} = \bar{v}_c + \frac{1 - 1/\psi}{1-\rho k_c} x_t, \tag{6}$$

$$r_{d,t+1} = \bar{r}_d + \frac{1}{\psi} x_t + k_d \frac{\lambda - 1/\psi}{1-\rho k_d} \epsilon_{x,t+1} + \epsilon_{d,t+1}, \tag{7}$$

$$r_{f,t} = \bar{r}_f + \frac{1}{\psi} x_t. \tag{4}$$

Interpretation. The long-run state x_t moves price-dividend ratios, the risk-free rate, equity returns, and the pricing kernel all at once. This is why it becomes a unifying state variable for international asset prices.

3.5 Exchange-rate surprise decomposition

The innovation in exchange-rate growth is

$$\Delta e_{t+1} - E_t[\Delta e_{t+1}] = k_c \frac{1-\gamma\psi}{\psi(1-\rho k_c)} (\epsilon^*_{x,t+1} - \epsilon_{x,t+1}) - \gamma(\epsilon^*_{c,t+1} - \epsilon_{c,t+1}). \tag{5}$$

Interpretation. This equation is the paper's central mechanism:

- short-run relative consumption shocks move exchange rates through the second term;
- long-run relative growth news moves exchange rates through the first term;
- the long-run channel matters only when risk aversion and intertemporal substitution are disentangled, that is, when $\gamma \neq 1/\psi$;
- and the stronger the persistence ρ, the larger the price effect of long-run news.

3.6 Monte Carlo detectability exercise

To show why long-run risks are hard to identify from consumption data alone, the paper simulates

$$\Delta c_t = x_{t-1} + \sigma \epsilon_{c,t}, \tag{8}$$

$$x_t = P x_{t-1} + J \sigma R^{1/2} \epsilon_{x,t}. \tag{6}$$

Here P collects the persistence parameters and R is the cross-country correlation matrix of the predictable components.

Interpretation. The Monte Carlo exercise is not part of the asset-pricing model itself. It is an identification argument: even if long-run risk is truly there, the sample may be too short to estimate it sharply from consumption data alone.

3.7 Predictive regressions for consumption growth

For country $i \in \{US, UK\}$, the paper estimates three specifications:

$$\Delta c_t^i = \beta_1^i pd_{t-1}^i + \epsilon_{c,1,t}^i, \tag{7}$$

$$\Delta c_t^i = \beta_2^i \left[pd_{t-1}^i, r_{f,t-1}^i\right] + \epsilon_{c,2,t}^i, \tag{8}$$

$$\Delta c_t^i = \beta_3^i \left[pd_{t-1}^i, r_{f,t-1}^i, \Delta c_{t-1}^i, cy_{t-1}^i, default_{t-1}^i\right] + \epsilon_{c,3,t}^i. \tag{9}$$

Interpretation.

- Equation (7) uses only the price-dividend ratio.
- Equation (8) adds the risk-free rate, which is also informative about the long-run state in the model.
- Equation (9) is the richest specification and uses the wider predictive set advocated in the long-run-risk literature.

The fitted values from these regressions are treated as empirical measures of the long-run component.

3.8 Time-varying correlation and volatility measures

To connect long-run-risk comovement to exchange-rate volatility, the paper constructs smoothed correlations and variances using exponential smoothing. Writing the demeaned fitted long-run-risk measures as $\tilde{x}_t^{US}$ and $\tilde{x}_t^{UK}$,

$$q_{12,t} = \alpha q_{12,t-1} + (1-\alpha)\tilde{x}_t^{US}\tilde{x}_t^{UK}, \tag{9}$$

$$q_{1,t} = \alpha q_{1,t-1} + (1-\alpha)(\tilde{x}_t^{US})^2, \tag{10}$$

$$q_{2,t} = \alpha q_{2,t-1} + (1-\alpha)(\tilde{x}_t^{UK})^2, \tag{11}$$

$$\rho_t^h = \frac{q_{12,t}}{\sqrt{q_{1,t}q_{2,t}}}. \tag{12}$$

Exchange-rate volatility is smoothed similarly with a decay parameter set to 0.95.

Interpretation. This turns the theory's cross-country long-run-risk correlation into a time series that can be compared directly with the time series of exchange-rate volatility.

3.9 Euler-equation GMM for consumption-based tests

Using the estimated long-run components, the paper imposes moment restrictions from the Euler equations for:

- domestic equity returns and risk-free rates,
- foreign equity returns and risk-free rates expressed in local units,
- the first two moments of exchange-rate growth,
- and the covariance between exchange-rate growth and cross-country consumption-growth differentials.

In one set of estimations, the time series of long-run risk is taken as given ("conditional estimation"). In a second set, preference parameters and the long-run-risk dynamics are jointly estimated.

Interpretation. This is the main empirical test of whether the mechanism can jointly rationalize asset prices and exchange-rate moments, not just predictive regressions.

3.10 Dividend predictive system

To bring cash-flow dynamics and return moments directly into the estimation, the paper jointly estimates the restricted system

$$\Delta c_t^i = \beta_0^i \Delta c_{t-1}^i + \beta_1^i pd_{t-1}^i + \beta_2^i cy_{t-1}^i + \beta_3^i default_{t-1}^i + \beta_4^i r_{f,t-1}^i + \epsilon_{c,t}^i, \tag{13}$$

$$\Delta d_t^i = \lambda^i \big(\beta_0^i \Delta c_{t-1}^i + \beta_1^i pd_{t-1}^i + \beta_2^i cy_{t-1}^i + \beta_3^i default_{t-1}^i + \beta_4^i r_{f,t-1}^i\big) + \epsilon_{d,t}^i, \tag{10}$$

for all $i \in \{US, UK\}$.

The unrestricted comparison system is

$$\Delta c_t^i = \beta_0^i \Delta c_{t-1}^i + \beta_1^i pd_{t-1}^i + \beta_2^i cy_{t-1}^i + \beta_3^i default_{t-1}^i + \beta_4^i r_{f,t-1}^i + \epsilon_{c,t}^i, \tag{14}$$

$$\Delta d_t^i = \phi_0^i \Delta c_{t-1}^i + \phi_1^i pd_{t-1}^i + \phi_2^i cy_{t-1}^i + \phi_3^i default_{t-1}^i + \phi_4^i r_{f,t-1}^i + \epsilon_{d,t}^i. \tag{11}$$

Interpretation. Equation (10) imposes the model's restriction that the same long-run component drives both consumption and dividend growth, up to a loading parameter λ. Equation (11) checks whether the data reject this restriction.

4 Identification and estimation logic (what makes the results credible)

4.1 What identifies the main mechanism

There is no external instrument in the baseline theoretical section. Instead, the logic is structural:

- the model says that exchange-rate volatility depends on the correlation of pricing kernels;
- recursive preferences imply that pricing kernels depend on long-run as well as short-run consumption news;
- highly correlated long-run components then make stochastic discount factors more correlated than raw annual consumption growth alone would suggest.

Interpretation. Identification in the theory comes from the structure of equation (5): long-run relative growth news moves exchange rates only when persistence is high and $\gamma \neq 1/\psi$.

4.2 Why consumption-only estimation is weak

The paper explicitly shows that a Kalman-filter likelihood based only on consumption data is not informative enough to pin down long-run components sharply in realistic samples. The Monte Carlo exercise makes this point quantitatively:

- with one consumption series, rejecting the random-walk benchmark is almost impossible;
- even with two or five countries, identification remains difficult unless the predictive components are nearly perfectly correlated;
- around 80 annual observations for both the United States and the United Kingdom is only barely enough even under favorable conditions.

Interpretation. This weak identification is not a failure of the model. It is the reason the empirical strategy turns to financial-price information.

4.3 Why financial predictors help

Price-dividend ratios, risk-free rates, consumption-output ratios, and default premia are forward-looking variables. If the economy contains a persistent expected-growth component, these variables should reflect it even when realized consumption growth looks close to a random walk.

Interpretation. The paper's empirical innovation is to treat asset prices as noisy but informative sufficient statistics for low-frequency growth news.

4.4 Why the U.S.-U.K. pair and the post-1971 window matter

The post-Bretton Woods sample is the main testing ground because:

- exchange-rate volatility changes dramatically once nominal pegs disappear,
- the degree of U.S.-U.K. financial openness is much higher after the late 1960s,
- and the model is meant to describe integrated asset markets rather than segmented ones.

Interpretation. Restricting attention to 1971–2006 is not cherry-picking. It aligns the empirical environment with the frictionless international-asset-market structure of the model.

4.5 Why the time-varying correlation-volatility fact is powerful

A major empirical implication is that when U.S. and U.K. long-run growth prospects become more similar, exchange-rate volatility should fall. The paper documents exactly this negative relation in both regression form (Table 5) and figure form (Figure 4).

Interpretation. This is strong evidence because it is not just one average moment; it is a time-varying comovement predicted directly by the mechanism.

4.6 Why joint estimation with returns strengthens the case

The conditional GMM exercises already show that the model needs moderate risk aversion and an elasticity of substitution above one. The joint estimations go further by forcing the data on returns, exchange rates, and the predictive process of consumption and dividends to fit simultaneously.

Interpretation. If the model only fit exchange rates in isolation, the result would be much less persuasive. The paper's credibility comes from matching several sets of moments at once.

4.7 What the paper can and cannot distinguish

- It shows that highly correlated long-run growth news can reconcile international prices and quantities, but it does not derive those long-run components from a fully specified production economy.
- It studies the correlation of pricing kernels, not a full theory of risk sharing with nontraded goods and home bias in consumption baskets.
- It abstracts from stochastic volatility, so it cannot explain the empirical failure of uncovered interest parity or the detailed dynamics of currency excess returns.
- The predictive-regression approach identifies a useful empirical proxy for long-run risk, but it is not the only possible filter and depends on financial variables being informative about future growth.

Interpretation. The paper is a strong mechanism paper, not a final all-purpose international macro-finance model.

4.8 Relation to the literature

The paper sits between three strands of work:

- the Backus-Smith international risk-sharing literature,
- the long-run-risk literature of Bansal and Yaron,
- and the exchange-rate literature emphasizing the disconnect between international prices and quantities.

Interpretation. Its main contribution is to import the long-run-risk logic into the international setting and show that it can solve a puzzle that is hard for standard complete-markets models.

5 Tables and figures

5.1 Table 1: baseline calibration

TABLE 1
BASELINE CALIBRATION

Variable	Definition	Value
Ψ	Intertemporal elasticity of substitution	2
γ	Risk aversion	4.25
δ	Subjective discount factor	.998
μ_c	Average consumption growth	15×10^{-4}
ρ	Autoregressive coefficient of the long-run component x_t	.987
φ_e	Ratio of long-run shock and short-run shock volatilities	.048
σ	Standard deviation of the short-run shock to consumption	68×10^{-4}
μ_d	Average dividend growth	.0007
λ	Leverage	3.0
φ_d	Volatility ratio of short-run shocks to dividend and consumption growth	5.0
ρ_x^{hf}	Cross-country correlation of the long-run shock	1.0
ρ_c^{hf}	Cross-country correlation of the short-run shock to consumption	.3
ρ_d^{hf}	Cross-country correlation of the short-run shock to dividends	−.1

NOTE.—The countries share the same calibration. The model describes a monthly decision problem. The parameters of the laws of motion of consumption and dividend growth are set to reproduce the average behavior of consumption growth in the United States and in the United Kingdom. The cross-country correlations of the idiosyncratic shocks to consumption and shocks to the trend are chosen so as to obtain a correlation of consumption growths in the order of 0.3. The coefficient λ is set in such a way that the ratio $\sigma_{\Delta d}/\sigma_{\Delta c} = 4.86$, which is in the range (4, 8) estimated by Ludvigson, Lettau, and Wachter (2008). The cross-country correlation of the short-run shocks to dividends, ρ_d^{hf}, is set so as to achieve an almost zero correlation of dividend growths between the United States and the United Kingdom.

Read: The benchmark monthly calibration sets $\psi = 2$, $\gamma = 4.25$, $\delta = 0.998$, persistence $\rho = 0.987$, very small long-run shock volatility relative to short-run consumption shocks, and perfect cross-country correlation of long-run shocks. Short-run consumption-shock correlation is only 0.3.

Economic message. The calibration is deliberately asymmetric in frequencies: short-run quantities are only modestly correlated, but long-run growth news is almost perfectly shared. That is what gives the model room to match smooth exchange rates without requiring implausibly smooth current consumption differences.

5.2 Figures 1 and 2: why intertemporal substitution and risk aversion matter

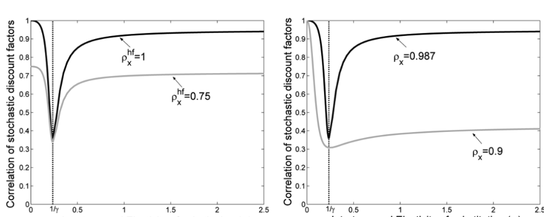

FIG. 1.—The role of intertemporal elasticity of substitution. In both panels, the dark line reports the correlation of stochastic discount factors when ψ changes. The grey line on the left panel is drawn for a smaller value of ρ_x^{hf}, everything else held equal; the grey line on the right panel is drawn for a lower value of ρ_x.

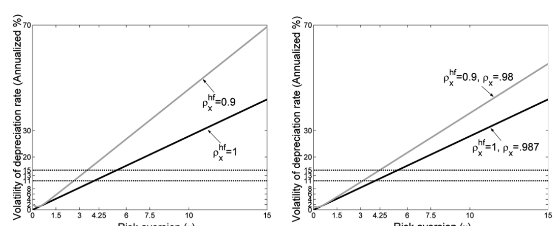

FIG. 2.—The role of risk aversion. In both panels, the dark line reports the volatility of the depreciation rate when γ changes. The grey line in the left panel is drawn for a smaller value of ρ_x^{hf}, everything else held equal; the grey line in the right panel is drawn for lower values of ρ_x and ρ_x^{hf}.

Read: Figure 1 shows how the correlation of stochastic discount factors changes with the intertemporal elasticity of substitution. The correlation is lowest near the CRRA case $\psi \approx 1/\gamma$ and rises sharply as ψ moves away from that value. Figure 2 shows how exchange-rate volatility rises with risk aversion and becomes counterfactually large if long-run shocks are less correlated or less persistent.

TABLE 2
KEY MOMENTS OF INTERNATIONAL RETURNS—SYMMETRIC CALIBRATION

Variable	Definition	United States	United Kingdom	Model
$\rho(m^h, m^f)$	Correlation of pricing kernels	. . .	. . .	.93
$\sigma(\Delta e)$	Volatility of FX growth	11.2		11.8
$E(\Delta e)$	Average FX growth	1.3		.0
$\rho(\Delta e_{t+1}, \Delta e_t)$	Autocorrelation of FX growth	.0		.0
$\rho(\Delta e_{t+1}, v_{d,t} - v^*_{d,t})$	Correlation of FX growth and price-dividend ratios differences	.0		.0
$\rho(\Delta e, \Delta c - \Delta c^*)$	Correlation of FX growth and consumption growth differentials	.15		.80
$\rho(v_d, v_{d,t-1})$	Autocorrelation of price-dividend ratio	.89	.82	.92
$\sigma(v_d)$	Volatility of price-dividend ratio	31.2	32.2	24.6
$E(v_d)$	Average price-dividend ratio	3.3	2.9	2.9
$E(r_d - r_f)$	Average excess return	5.5	6.5	5.3
$\sigma(r_d - r_f)$	Volatility of excess return	17.1	22.8	19.1
$E(r_f)$	Average risk-free rate	1.5	1.6	1.3
$\sigma(r_f)$	Volatility of risk-free rate	1.5	2.9	1.2
$\rho(r_d - r_f, r^*_d - r^*_f)$	Correlation of excess returns	.67		.60
$\rho(v_d, v^*_d)$	Correlation of price-dividend ratios	.77		.93
$\rho(\Delta d, \Delta d^*)$	Correlation of dividend growth	−.03		−.07
$\rho(r_f, r^*_f)$	Correlation of risk-free rates	.65		1.00
$E(r_d - r_f + \Delta e)/\sigma(r_d - r_f + \Delta e)$	Foreign Sharpe ratio (domestic units)	.20	.31	.23
$\rho(r_d - r_f, r^*_d - r^*_f + \Delta e)$	Correlation of excess returns, domestic units	.63	.60	.54
$\rho(r_d - r_f, \Delta e)$	Correlation of excess returns, local units, and FX growth	.04	−.03	.00
$\sigma(\Delta c)$	Volatility of consumption growth	1.4	2.9	2.5
$\sigma^2(x)/\sigma^2(\Delta c) \times 100$	Share of predictable consumption variance	. . .	. . .	8.2
$\sigma(\Delta d)$	Volatility of dividend growth	16.9	6.9	11.9
$\rho(\Delta c, \Delta c^*)$	Correlation of consumption growth	.28		.35

NOTE.—Data are quarterly real per capita and are from 1971:01 to 1998:04, as in Brandt et al. (2006). Means and variances are annualized and multiplied by 100. All variables are in log units and are measured in terms of local real consumption, "local units," unless noted otherwise. Domestic (foreign) units stands for variables measured in terms of the same real domestic (foreign) consumption good. FX denotes log exchange rate growth. The model calibration is reported in table 1. All simulated variables are time aggregated to an annual frequency.

Economic message. These two figures visualize the paper's central parameter logic. It is not enough to have recursive preferences in the abstract. The model needs both: (i) sensitivity to long-run risk, and (ii) highly persistent and highly correlated long-run components. Otherwise exchange rates become too volatile.

5.3 Table 2: symmetric calibration versus international moments

Read: Table 2 compares data moments for the United States and the United Kingdom with the model under the benchmark symmetric calibration. The model delivers pricing-kernel correlation around 0.93, exchange-rate volatility around 11.8%, excess-return correlation around 0.60, realistic price-dividend-ratio volatility and persistence, and sizable equity premia.

Economic message. The symmetric calibration already gets surprisingly far. It shows that the long-run-risk channel can simultaneously match exchange-rate volatility and strong international stock-return comovement, which is difficult in standard models. The main remaining weakness at this stage is the too-high Backus-Smith correlation and the overly high correlation of risk-free rates.

5.4 Table 3 and Table 4: detecting long-run components

Read: Table 3 is the Monte Carlo identification exercise. It shows that even with long samples, consumption data alone often cannot reject a random walk unless the predictive components are extremely correlated. Table 4 then reports the actual predictive-regression evidence: the long-run component exists, explains roughly 7% to 32% of annual consumption-growth variance, is highly persistent, and is strongly correlated across the United States and the United Kingdom.

TABLE 3
Filtering Predictive Components

$\rho(x, x^*)$	Number of Δc's	Number of x's	Sample Size: 50	80	100	200
1	1	1	(.00, 2.10)	(.00, 1.47)	(.00, 1.23)	(.00, .74)
	2	1	(.00, .69)	(.06, .66)	(.09, .63)	(.17, .52)
	5	1	(.13, .55)	(.18, .50)	(.20, .47)	(.24, .43)
.95	2	2	(.00, 1.10)	(.00, .74)	(.00, .67)	(.09, .55)
	5	5	(.00, .58)	(.11, .55)	(.14, .52)	(.21, .44)
.9	2	2	(.00, 1.15)	(.00, .78)	(.00, .68)	(.13, .58)
	5	5	(.00, .61)	(.08, .56)	(.13, .53)	(.21, .45)

Note.—Each column reports the 95 percent confidence interval for the estimated φ_e parameter for simulated samples of increasing size. The true value of φ_e is 0.34.

TABLE 4
Predictive Regressions: The Three Ingredients

	F-Statistic: United States (1)	F-Statistic: United Kingdom (2)	R^2: United States (3)	R^2: United Kingdom (4)	ρ_x: United States (5)	ρ_x: United Kingdom (6)	corr(x^{US}, x^{UK}) (7)
Price-dividend and risk-free	7.19	3.47	.16	.08	.768	.787	.579
	(.00)	(.03)			(.076)	(.074)	[.137, .896]
All predictive variables	9.07	6.85	.39	.27	.672	.759	.758
	(.00)	(.00)			(.074)	(.074)	[.531, .922]
Price-dividend only	7.29	6.01	.08	.07	.885	.726	.849
	(.00)	(.01)			(.065)	(.099)	[.762, .941]
VAR	11.71	4.20	.32	.15	.869	.765	.717
	(.00)	(.00)			(.058)	(.075)	[.172, .953]

Note.—Data are annual and go from 1930 to 2006. Columns 1 and 2 report the F-statistics and the associated p-values for the test that all the coefficients of the predictive regressions are equal to zero for the corresponding row model for the United States and the United Kingdom. Columns 3 and 4 report the R^2 for the predictive regressions for the United States and the United Kingdom. Columns 5 and 6 report the estimated autoregressive coefficients of the predictive components of U.S. and U.K. consumption growth. Column 7 reports the estimated cross-country correlation of the predictive components along with their bootstrapped confidence interval for the 1971–2006 period.

Economic message. The two tables are meant to be read together. Table 3 says, "do not expect raw consumption data to identify this state cleanly." Table 4 says, "once you let financial variables help, the state appears clearly." This is the paper's main empirical identification move.

5.5 Figure 3, Table 5, and Figure 4: time-varying long-run-risk correlation and FX volatility

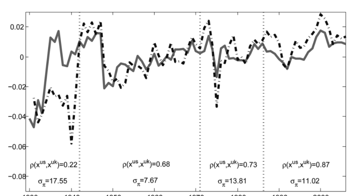

Fig. 3.—Time series of long-run risks in the United States and in the United Kingdom. We plot the fitted values of regression (9) for the United States (solid line) and the United Kingdom (dash-dot line). The numbers between the vertical dotted lines report the correlation of these projections and the volatility of the growth of the exchange rate in the corresponding subsample.

TABLE 5
Correlations and Volatilities

	R^2 (1)	β (2)	Var $(x^{US} - x^{UK})$/ Var (Δe) (3)
Price-dividend and risk-free	.929	−2.456	.007
		[.013]	
All predictive variables	.610	−9.809	.004
		[.077]	
Price-dividend only	.792	−5.996	.003
		[.014]	
VAR	.924	−7.419	.061
		[.092]	

Note.—Columns 1 and 2 report the R^2 and the slope coefficient in the regression of the exchange rate's volatility on the cross-country correlation of long-run risks, respectively. The numbers in parentheses are the bootstrapped probabilities of observing a nonnegative slope. Column 3 displays the faction of exchange rate growth variance that is accounted for by the variance of the difference of the predictive components of consumption growth. The sample is 1971–2006.

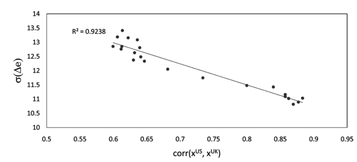

Fig. 4.—Correlations of long-run risks and volatilities of exchange rate movements. The correlations of long-run risks are measured as the expected discounted sum of consumption growth rates obtained from the VAR against the volatility of the growth of the exchange rate between U.S. dollars and British pounds. The sample used in the estimation is 1971–2006.

Read: Figure 3 plots the estimated long-run components for the United States and the United Kingdom and highlights that their correlation rises over time. Table 5 runs regressions of exchange-rate volatility on the cross-country correlation of long-run risks and finds a robustly negative slope across alternative measures. Figure 4 shows this relationship in a scatter plot, with a tight downward-sloping cloud.

Economic message. This is some of the paper's cleanest evidence. When long-run growth prospects across the two countries move more together, exchange-rate volatility falls. That is exactly what the mechanism predicts, and it is hard to generate from a model that only emphasizes short-run consumption differences.

5.6 Table 6: Euler-equation GMM on consumption moments

Read: Table 6 estimates preference parameters and, in the joint specifications, the long-run-risk process itself using Euler-equation restrictions together with exchange-rate moments. In the conditional estimates, the intertemporal elasticity falls from about 4.1 to about 1.3 depending on the predictor set, while risk aversion is around 3 to 4.4. In the joint estimates, ψ is around 2 to 2.8, γ

TABLE 6

Euler Equations' Generalized Method of Moments (GMM)

	Conditional Estimation			Joint Estimation		
	P/D (1)	P/D, R_f (2)	All (3)	P/D (4)	P/D, R_f (5)	All (6)
ψ	4.094	1.371	1.276	2.781	2.719	2.001
	[.398, 6.739]	[.391, 1.792]	[.404, 1.992]	[.243, 2.971]	[.288, 3.030]	[.274, 3.099]
γ	4.359	3.402	2.936	3.387	3.225	4.147
	[3.181, 9.575]	[2.282, 7.376]	[2.144, 6.339]	[1.014, 10.267]	[1.090, 9.509]	[1.167, 10.802]
ρ_x	. . .	. . .	. . .	.997	.997	.988
				[.427, .999]	[.413, .999]	[.662, .999]
$\rho(x^{US}, x^{UK})$	. . .	. . .	. . .	.879	.867	.830
				[.618, .936]	[.582, .939]	[.471, .963]
Wald statistic p-value	. . .	. . .	. . .	.000	.000	.000
$\gamma - 1/\psi$ p-value	.000	.002	.008	.000	.001	.000
J-statistic p-value	.000	.000	.000	.035	.000	.011

Note.—Each column reports the parameters estimated using GMM on the 1971–2006 sample. All GMMs have the following set of moment conditions in common: Euler equations for domestic stock market returns and risk-free rates (4), Euler equations for foreign stock market returns and risk-free rates expressed in local units (4), first two moments of the exchange rates' growth (2), and the covariance between exchange rates' growth and cross-country consumption growth differences (1). The weighting matrix is the identity matrix. The GMMs in cols. 1–3 are conditional on the ordinary least squares estimates of the regression of consumption growth of the previous section. For the results reported in cols. 4–6, all models' parameters are estimated jointly by adding the appropriate set of orthogonality restrictions. The numbers is brackets below each estimate are bootstrapped 95 percent confidence intervals. Wald statistics test the null of no predictability in consumption growth rates in both countries. The row $\gamma - 1/\psi$ p-value reports the p-value for the one-tailed null $H_0: \gamma \leq 1/\psi$.

is around 3.2 to 4.1, persistence is near one, and the cross-country correlation of long-run risks is around 0.83 to 0.88.

Economic message. The table says two important things. First, the data favor $\psi > 1$ and reject the null $\gamma \leq 1/\psi$, so agents prefer early resolution of uncertainty. Second, once the long-run state is estimated jointly, the model is no longer decisively rejected in its richer versions. Exchange-rate data are doing a lot of identification work here.

5.7 Table 7 and Table 8: dividends and joint estimation with returns

TABLE 7

Predictive Regressions on Dividends

	Δc_{t-1} (1)	pd_{t-1} (2)	cy_{t-1} (3)	default_{t-1} (4)	$r_{f,t-1}$ (5)	λ (6)	$R^2_{\Delta d}$ (7)	Likelihood Ratio Test (8)	ρ_x (9)
United States	.455	.011	−.063	1.289	−.045	2.653	.071	13.28	.673
	(.150)	(.006)	(.043)	(.739)	(.134)	(1.052)		(.010)	(.074)
	. . .	.016	. . .	. . .	−.299	5.492	.159	.420	.768
		(.005)			(.095)	(1.720)		(.520)	(.076)
	. . .	.014	. . .	. . .	. . .	5.965	.113	. . .	.887
		(.006)				(2.252)			(.056)
United Kingdom	.268	.032	−.088	. . .	−.226	1.666	.129	1.460	.761
	(.128)	(.020)	(.030)		(.531)	(.580)		(.691)	(.074)
	. . .	.031	. . .	. . .	.183	2.097	.063	.520	.787
		(.024)			(.254)	(1.249)		(.471)	(.073)
	. . .	.033	. . .	. . .	. . .	2.297	.067	. . .	.737
		(.024)				(1.360)			(.081)

Note.—In cols. 1–6, we report the maximum likelihood estimates of the parameters in (10). Newey-West adjusted standard errors are reported in parentheses. Column 7 shows the implied R^2 for dividend growth. Column 8 reports the likelihood ratio tests (p-values in parentheses) with respect to the unrestricted system (11). Column 9 reports the estimates of the persistence of the predictable component. Data are annual and demeaned. The sample ranges from 1929 to 2006.

TABLE 8

GMM Estimation with Dividends

	Conditional Estimation			Joint Estimation		
	P/D (1)	P/D, R_f (2)	All (3)	P/D (4)	P/D, R_f (5)	All (6)
ψ	2.188	2.068	1.893	2.095	2.106	1.884
	[1.319, 2.473]	[1.518, 2.240]	[1.410, 2.153]	[.936, 2.909]	[.905, 3.005]	[.880, 3.217]
γ	5.066	3.772	2.916	2.741	2.843	1.988
	[3.636, 8.827]	[2.647, 7.721]	[2.398, 4.818]	[1.020, 4.765]	[1.013, 5.666]	[.995, 4.287]
ρ_x	. . .	. . .	. . .	.981	.976	.988
				[.747, .999]	[.711, .999]	[.694, .999]
$\rho(x^{US}, x^{UK})$	. . .	. . .	. . .	.879	.879	.941
				[.645, .938]	[.588, .933]	[.467, .965]
Wald-statistic p-value	. . .	. . .	. . .	.000	.000	.000
$\gamma - 1/\psi$ p-value	.000	.000	.000	.000	.002	.008
J-statistic p-value	.000	.000	.000	.160	.023	.240

Note.—Each column reports the parameters estimated using GMM on the 1971–2006 sample. Data are annual. All GMMs have the following set of moment conditions in common: first moment for stock market returns and risk-free rate (4), first two moments of the exchange rates' growth (2), and the covariance between exchange rates' growth and cross-country consumption growth differences (1). The weighting matrix is the identity matrix. The first three GMMs are conditional on the ordinary least squares estimates reported in table 7. In cols. 4–6 we report results of the joint GMM estimation of (4) and (10). The appropriate orthogonality restrictions are added. The numbers in brackets below each estimate are bootstrapped 95 percent confidence intervals. Our Wald statistics are computed imposing all the restrictions required to test the null of no predictability in consumption and dividend growth rates in both countries. The row $\gamma - 1/\psi$ p-value reports the p-value for the one-tailed null $H_0: \gamma \leq 1/\psi$.

Read: Table 7 shows that dividend growth is also predictable from the same financial variables and that the restricted system linking dividend predictability to the same long-run component as consumption is not rejected. Table 8 then brings dividends and returns directly into the GMM estimation. In the joint specifications, ψ remains close to 2, γ stays moderate, persistence remains very high, and the J-test p-values improve to economically comfortable levels such as 0.160 and 0.240.

Economic message. These tables strengthen the overall argument by extending it from consumption and exchange rates to dividend cash flows and return moments. The point is not just that a latent long-run component exists; it is that the same component helps explain both macro quantities and financial prices.

TABLE 9
Estimated Key Moments of International Returns

	$\sigma(\Delta e)$ (1)	$AC_1(\Delta e)$ (2)	$\rho(\Delta e, \Delta c - \Delta c^*)$ (3)	$\rho(r_d - r_f, r_d^* - r_f^*)$ (4)	$E[r_d - r_f]$ (5)	Sharpe Ratio (6)	$E[r_f]$ (7)
United States:							
Data	.120	.006	.003	.756	5.083	.297	1.009
Model	.166	−.130	.224	.721	4.855	.151	1.200
United Kingdom:							
Data	. . .	. . .	. . .	. . .	5.356	.207	1.525
Model	. . .	. . .	. . .	. . .	4.511	.145	1.469

Note.—Data are annual real per capita and are from 1971 to 2006. The model is calibrated according to col. 6 of table 8. Means are denoted by $E[\cdot]$ and multiplied by 100. Standard deviation and contemporaneous correlation are denoted by $\sigma(\cdot)$ and $\rho(\cdot,\cdot)$, respectively. In col. 2, AC_1 denotes first-order autocorrelation. Columns 1–4 refer to international moments that are identical across the United States and the United Kingdom.

5.8 Table 9: final model-implied international moments

Read: Table 9 reports key moments implied by the fully estimated model. Exchange-rate volatility is about 16.6%, close to the data; exchange-rate autocorrelation is close to zero; the cross-country correlation of equity returns is about 0.72 versus 0.76 in the data; the Backus-Smith correlation drops sharply relative to the simple benchmark; and the model still generates sizable equity premia with low risk-free rates.

Economic message. This is the paper's final scorecard. The estimated model does not hit every moment exactly, but it comes much closer than the standard benchmark on the moments that define the international prices-versus-quantities puzzle.

6 Short summary

- Standard international complete-markets models struggle because weakly correlated consumption growth implies either implausibly volatile exchange rates or implausibly high risk aversion.
- Adding Epstein-Zin preferences and a persistent long-run component to consumption growth changes the picture because future growth news affects current marginal utility.
- If the long-run components are highly correlated across countries, stochastic discount factors can be highly correlated even when current consumption growth is not.
- Consumption data alone cannot identify these long-run components well, so the paper uses predictive information from asset prices to measure them.
- In U.S.-U.K. data, the estimated long-run components are persistent, strongly correlated, and more correlated precisely when exchange rates are less volatile.
- The resulting model goes a long way toward reconciling international exchange rates, stock returns, price-dividend ratios, and equity premia in one frictionless framework.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that a frictionless international asset-pricing model can reconcile a large part of the "international prices versus quantities" puzzle once it incorporates two ingredients: recursive preferences and a highly persistent, strongly cross-country-correlated long-run component in consumption growth.

7.2 Economic intuition

The economic intuition is simple but powerful. Exchange rates do not only reflect who had a good current year in consumption terms. They also reflect how the market values the whole future path of relative consumption. If long-run growth news is mostly shared across countries, then a lot of that future news cancels out in relative prices, making exchange rates much smoother than short-run consumption data alone would imply.

7.3 Takeaway

This paper is one of the clearest demonstrations that long-run risks are not only a domestic asset-pricing device. They also matter centrally for international macro-finance. The broader lesson is that exchange-rate dynamics may look puzzling only if we focus too narrowly on high-frequency consumption movements and ignore what asset prices know about the long run.